The EM chapters, re-examined

Confirmatory evidence for Chapters 8–11

Internal report for discussion with Jerome and Marc · 18 September 2026

Second revision. Changes against the 16 September draft are listed on this page.

This report states what sixteen-seed confirmatory runs, executed on the Bocconi cluster over the past four days, say about the chapters from EM onwards. Three of the four chapters survive intact; one conclusion reverses; one comparison the thesis explicitly declared out of reach has now been made. Section 1 is the short version, Section 2 states what we are doing and why it is trustworthy, Sections 3–6 go chapter by chapter, Section 8 reports what four further runs say about how far the results travel, and Section 10 lists the decisions that need taking.

What changed since the 16 September draft. Four runs have reported. Two of them confirm and extend; two of them require corrections that are made here and not buried.

- The misspecification sweep was repeated at sixteen seeds. **One of its four conclusions does not survive**: long-range coupling was reported as reversing the ordering near $\gamma \approx 0.4$; at sixteen seeds it does not reverse anywhere in the swept range. The remaining three mechanisms are unchanged and two of them now have tighter brackets (Section 7).
- The capacity contrast was re-run across five correlations and three chain lengths and holds at all fifteen, with the apparatus checking itself at every one (Section 8.1).
- A sweep of the quadrature found that the reference grid's accuracy is bounded by the *domain*, not by the number of points, and that one configuration the phase appendix reports — the heavy-tailed family at zero correlation — lies outside the validated envelope (Section 8.2). Those entries are flagged and not cited pending a recomputation.
- An audit of the EM iteration budget across all runs **flags one reported claim**: the statement that the estimator leaves the information bound at the two highest noise levels is confounded with the ladder running out there (Section 8.3). The capacity conclusion is cleared by the same audit.

1 Summary

1. **The capacity conclusion of Chapter 10 reverses.** It was an artefact of a stopping rule that fired for the single-Gaussian arm and for no other. Under a protocol that selects the checkpoint identically for every capacity, the single Gaussian sits exactly on the second-order floor at every sample size (0.167) and every richer capacity goes far below it (0.023 at $M = 2048$), favouring the mixture in 16 of 16 seeds at every size.

2. **The two rates of Chapter 9 are a property of the channel, not of the optimiser.** The EM estimator attains the Cramér–Rao bound of the noised likelihood to within a factor 1.02 on average. The shape coordinate is not slow because EM is slow; the data stop carrying it. Its information bound crosses the coordinate's own range at $t = 0.98$, and that is precisely where the fit collapses to a Gaussian innovation.

3. **The comparison of Chapter 10 now runs at a matched information interface.** The thesis attached a scope condition saying it could not do this. Given the same single noised view per sequence that the estimator sees, the best network reaches 0.155 against the estimator's 0.023: a paired factor 7.3, interval $[6.2, 8.7]$. The unmatched comparison, at 2.83×, *understates* the gap.

4. **Locality is a ceiling, not only a prior.** The radius-4 window head is the best network at $M = 32$ and the worst of the five weight-shared architectures at $M = 2048$, moving only $0.121 \to 0.118$ as data quadruples. A bounded receptive field cannot represent a message that has crossed the chain — which is exactly what the chain's score is.

5. **Score accuracy transfers to generation, and the second-order score fails there visibly.** Driven through the same reverse SDE on identical Brownian paths over sixteen seeds, the fitted kernel produces sequences whose innovation excess kurtosis is 1.80 against the reference's 1.94, at a log-likelihood deficit of +0.09 nats per sequence against the arm driven by the exact score (interval [+0.00, +0.19]). The covariance-matched score gives 0.00 — an essentially Gaussian chain — at a deficit of +1.70, and the isotropic score +46.2. An earlier four-seed reading had the fitted kernel level with the exact score; sixteen seeds show a small, real deficit (Section 7).

6. **A claim of ours was refuted by our own data, and is recorded as such.** "The learned networks sit at or above the second-order floor" had been drafted into the new abstract. At $M = 2048$ five of six architectures sit below it. The defensible statement is about the information interface, not about a barrier the networks cannot pass.

7. **The capacity result travels, and checks the apparatus while it does.** Across five correlations and three chain lengths the mixture beats the Gaussian kernel in all fifteen cells, with every interval excluding zero and the margin shrinking smoothly with ρ as the marginal's excess kurtosis does. In the same fifteen cells the $C = 1$ arm lands on the LMMSE of the true covariance to 0.01, as Proposition 1 requires — a simultaneous check on the grid, the message passing, the EM and the scoring, passed at fifteen independent configurations.

8. **Two corrections, both found by us and both made here.** Long-range coupling does *not* reverse the ordering at sixteen seeds, though the eight-seed draft said it did; and the statement that the estimator leaves the information bound at the two highest noise levels is confounded with the EM ladder running out there. The first is settled and rewritten, the second is flagged with a control queued. A third finding of the same kind is about our instrument rather than our results: the reference grid's accuracy is limited by the width of its domain rather than by the number of points, and one configuration the phase appendix reports lies outside the validated envelope.

2 What we are doing, and why the numbers can be trusted

Three changes to how the work is run account for why these results settle questions the thesis left open.

One frozen protocol, one varied factor. Everything the arms share is now a single frozen object: the bundle sizes and their random streams, the sizes $M \in \{32, 128, 512, 2048\}$, the sixteen confirmatory seeds and four development seeds, the EM iteration ladder and its initialisation, the network training recipe and screen budget, the estimand, and the scoring regions. Each experiment varies exactly one declared factor against that background, so a difference between two arms is attributable to that factor. Two arms that differ in more than one thing are not compared; where a comparison genuinely needs two factors to move, the intermediate arm is run as well.

Pairing by construction. Every row records the SHA-256 of the validation and test bundles it was scored on. Rows from different experiments are joined on (seed, size, bundle hash), and

a generator that is handed rows with mismatched hashes refuses to emit a number. This is what makes "EM–BP against the best network" a paired comparison across two experiments rather than two averages.

Frozen outputs. A result becomes citable only when its output directory is pinned: every CSV and provenance file hashed into the registry entry together with the commit that produced it. After that, any byte change fails the build, and the manuscript refuses to compile if a generated number disagrees with the artefact it came from. A re-run is a new artefact under a new key, never an overwrite. Confirmatory seeds are declared before the run and never used for selection; development seeds never enter a reported number.

Amendments are recorded, not absorbed. Two design changes were made this week before the runs that used them, and both are in the registry with timestamps: the EM ladder was extended from 400 to 1200 iterations because 400 was binding at the largest sample size, and the largest strength of the site-varying misspecification mechanism was removed because it is not a valid chain — at $\rho = 0.85$ its site coefficients exceed 1 and the implied covariance is not positive definite. No number from the invalid point exists.

3 Chapter 8 — EM for the parameters: holds

Nothing in the derivation needed revision. Fisher's identity still gives the exact gradient of the discretised marginal log-likelihood from one forward–backward pass, and the finite-difference check agrees to about 10^{-9}. Two things were added.

The algebra is now machine-checked where checking is cheap. A Lean 4 module builds against Mathlib with no `sorry` and no axiom, covering four constants: the Hermite attenuation for degrees up to four, the fourth-cumulant difference under variance matching, the factor $\frac{1}{2}$ in the Gaussian information rate, and the coefficient $\kappa_4/6$ in the score gap. It verifies the algebraic skeleton and not the analysis: the remainder bounds and the measure-level statements remain informal, and the write-up says so in those words rather than claiming more.

The confirmatory ladder was extended to 1200 iterations. At $M = 2048$ the richer capacities were still selecting the top of the 400-iteration ladder; they still select the top of the extended one in 10 of 16 seeds, so the errors reported for those arms in Table 1 remain upper bounds. That can only understate the effect described there.

4 Chapter 9 — the two rates: same fact, better reason

The chapter explains the asymmetry through EM's fixed-point map: the Jacobian is the missing information and the shape coordinate has the smaller eigenvalue, so the fitted innovation shape settles at a median of 229 updates against 80 for the correlation. That measurement stands, and the stopping-rule warning built on it stands with it.

What is new is that the asymmetry is no longer only a statement about the optimiser. Two results bracket it.

The channel, from theory. Under the Ornstein–Uhlenbeck channel every Hermite mode of degree m is attenuated exactly by e^{-mt}, so the Fisher information one noised sequence carries about order-m structure decays as e^{-2mt}: correlation as e^{-4t}, non-Gaussian shape as e^{-8t}. Computed on the chain by exact inference the measured slopes are -3.97 and -8.01 against the predicted -4 and -8, and the constant in the score gap between the true score and that of the covariance-matched Gaussian is reproduced to within one per cent.

The estimator, from measurement. Fitting the transition law from sequences observed at a single level t ($M = 512$, 16 seeds, one noised view per sequence), the EM estimator sits *at* the Cramér–Rao bound of the noised likelihood: a ratio of 1.02 on average over the levels where either coordinate is informative, never worse than 1.18 for the correlation or 1.51 for the fourth cumulant.

Taken together these say the chapter's phenomenon belongs to the channel. If the estimator is efficient, the two rates cannot be blamed on EM. The bound for the shape coordinate grows as e^{4t} and crosses the coordinate's own range ($\kappa_4 = 3$) at $t = 0.98$; the fit collapses to a Gaussian innovation at exactly that level, while the correlation estimate is still within 0.009 of the truth and survives past the crossover $t_* = 1.17$.

One limitation, stated plainly. The corollary predicts errors growing as e^{2t} and e^{4t}, and this run cannot test those exponents. At $\rho = 0.85$ the shape estimate saturates before the crossover, so the asymptotic window does not exist at this configuration. Fitting slopes through a saturated estimate would have returned 3.64 and 0.09 against predictions of 2 and 4; the analysis therefore reports no slope, and a companion run at $\rho = 0.5$ with $M = 2048$ — where the crossover falls at 0.55 and the window is reachable — is registered and queued.

5 Chapter 10 — the comparison: one reversal, one replacement, two new facts

5.1 The capacity sweep reverses

The chapter concluded that no mixture capacity improves on a single Gaussian innovation, and flagged its own caveat in the accompanying figure: the share of fits stopped at the iteration cap, and with a narrowest component below the grid's resolution floor, both climbed steeply with C. That caveat was the entire effect. In the archival sweep the likelihood tolerance fired in 31 of 32 single-Gaussian cells at a median of 71 iterations — close to where a validation protocol would stop — and in none of the 96 cells with $C \geq 4$, which therefore reported checkpoints an order of magnitude past their optimum. "The single Gaussian is as good" was "the single Gaussian is the only arm that stopped in the right place".

Table 1 re-runs the sweep with the checkpoint selected on a per-seed validation bundle identically for every capacity.

Table 1: Capacity under equal stopping. Schedule-pooled relative score error on all sites, mean over 16 seeds, every arm on identical held-out bundles. The last column is the same capacity fitted to the innovation law itself — the C-component mixture closest in Kullback–Leibler divergence — an infinite-data reference for the family rather than a bound. The isotropic score is at 0.669 at every size.

	sequences M				
kernel	32	128	512	2048	$M = \infty$
LMMSE floor	0.167	0.167	0.167	0.167	0.167
$C = 1$	0.174	0.168	0.168	0.167	0.167
$C = 2$	0.084	0.050	0.035	0.028	0.039
$C = 4$	0.093	0.053	0.037	0.025	0.0057
$C = 8$	0.099	0.055	0.036	0.023	0.0022
$C = 16$	0.100	0.056	0.036	0.022	0.0007

Three readings follow. The single-Gaussian kernel lands on the second-order floor at every sample size, which is Proposition 1 read as a statement about an estimator rather than about a projection. Every richer capacity goes below that floor, by a factor 7.9 at the largest size; the

paired difference between $C = 1$ and $C = 8$ is 0.144 with bootstrap interval $[0.141, 0.147]$ and favours the mixture in 16 of 16 seeds. And the best capacity rises with the data, from $C = 2$ at $M = 32$ to $C = 16$ at $M = 2048$ — the ordinary variance-to-bias trade, visible now that the stopping rule no longer masks it.

The check that closes the loop. Two things changed at once between the thesis sweep and this one: the protocol, and the estimand (held-out log-evidence per edge there, relative score error here). Two simultaneous changes are not an explanation. A registered run now scores the *same frozen fits* in the thesis's own estimand, leaving the protocol as the only difference. A single-seed pass already shows the reversal — log-evidence per edge -1.3116 for $C = 8$ against -1.3146 for $C = 1$, with the true kernel at -1.3076 — and the sixteen-seed result is running.

5.2 The data-efficiency comparison is replaced

The chapter reports 6.6–10.9× against unstructured denoising score matching and 2.3–6.1× against a baseline given locality and weight sharing, with three scope conditions, the sharpest being that the arms are asymmetric in supplied information and that the comparison therefore does not rank learning algorithms at a matched interface.

Six architectures were screened under one recipe — 68 configurations, four development seeds, three restarts, 32,000 steps each — and the winner of each family was run on the sixteen confirmatory seeds (Table 2).

Table 2: Under one training recipe, 16 confirmatory seeds, identical held-out bundles. All-site schedule-pooled relative score error. The upper block is trained on clean sequences with a fresh corruption at every gradient step; the lower block sees what the estimator sees.

estimator	$M = 32$	$M = 2048$	information supplied
fully connected	0.517	0.208	clean, fresh corruption
window head ($r = 4$)	0.186	0.118	clean, fresh corruption
bidirectional GRU	0.201	0.060	clean, fresh corruption
transformer	0.231	0.063	clean, fresh corruption
1-D U-Net	0.227	0.066	clean, fresh corruption
dilated TCN	0.223	0.068	clean, fresh corruption
best matched-interface net	—	0.155	one noised view
EM–BP ($C = 8$)	0.099	0.023	one noised view
LMMSE floor	*0.167*	*0.167*	—

Two numbers replace the thesis's ranges. Against the best of six architectures under the standard recipe, the estimator is ahead by a paired factor 2.83, interval $[2.37, 3.42]$, in 16 of 16 seeds at $M = 2048$, and by 2.07 at $M = 32$. Against the best network trained at the estimator's own interface — one noised view per sequence, no clean chain, checkpoint selected on the implicit objective of a held-out noisy bundle and never on the true score — the factor is 7.3, interval $[6.2, 8.7]$. Implicit score matching on that bundle, which uses no clean target at all, lands at 0.165.

The second number is the one the thesis said was out of reach, and it is the stronger claim: the gap is not an artefact of giving the networks less, because when they are given the same, the gap widens.

It also follows that one sentence drafted for the new paper is false and has been withdrawn. "Every network sits at or above the second-order floor" holds only at the smallest sample size; at $M = 2048$ five of the six sit below it. Given enough clean sequences and fresh corruptions a neural denoiser plainly does learn fourth-order structure. The honest statement concerns the

information interface.

5.3 Two facts the thesis does not contain

Locality is a ceiling. The window head is the best network at $M = 32$ and the worst of the five weight-shared architectures at $M = 2048$, moving only from 0.121 to 0.118 as the data quadruples, against a standard error of 0.001, while the global architectures continue to 0.060. A radius-4 window cannot represent a message that has travelled the length of the chain, and the chain's exact score is precisely such a message. This connects the architecture comparison to the bulk-posterior analysis the chapter already contains.

Length extrapolation separates the arms cleanly. Fitted at $L = 32$ and evaluated against the exact score of longer chains (Table 3), the fitted kernel is flat: the law is local, so inference at any length is the same exact recursion. The convolutional and recurrent networks improve slightly, as they should, since longer chains have proportionally fewer boundary sites. The transformer with learned relative positions degrades by an order of magnitude — a failure of the positional encoding beyond its training length, not of the score it learned.

Table 3: Length extrapolation at $M = 2048$: pooled relative score error of estimators fitted at $L = 32$ and evaluated against the tree-exact score at longer chains, mean over 16 seeds.

estimator	$L = 32$	$L = 64$	$L = 128$	$L = 256$
EM–BP ($C = 8$)	0.023	0.023	0.023	0.023
bidirectional GRU	0.060	0.057	0.057	0.056
1-D U-Net	0.066	0.064	0.062	0.062
dilated TCN	0.068	0.066	0.065	0.064
window head	0.118	0.099	0.088	0.082
transformer	0.063	0.243	0.514	0.703

6 Chapter 11 — which answers change

RQ5, recovering an unknown transition. The answer holds and gains a sentence. The E-step is structurally exact, recovery is demonstrated numerically over the studied regime, and the uneven settling of the coordinates is real. We can now add that the estimator is statistically efficient, so the unevenness belongs to the channel and not to the algorithm. The stopping-rule warning becomes sharper rather than weaker: a rule read off the correlation trace certifies a kernel whose shape coordinate the data may no longer identify at all.

RQ6, what supplying the Markov assumption is worth. The answer changes in its numbers and in its scope. The ranges 6.6–10.9× and 2.3–6.1× are superseded by 2.83× against the best of six architectures under one recipe. The third scope condition — that the comparison does not rank methods at a matched information interface — can be removed, because that comparison now exists and gives 7.3×. The capacity subsection of the same chapter must be rewritten in the opposite direction.

The retired claim. The capacity result should be stated as retired rather than quietly dropped, together with the diagnosis of why it arose. The same applies to the claim of ours that our own screen refuted this week. The project keeps a negative-results record for exactly this, and both entries are in it.

7 Generation and the boundary of the advantage

Two results completed while this report was being written. Both were pre-registered and both address questions Chapter 10 raises but does not settle.

Does score accuracy transfer to generation? Every arm drives the same reverse SDE from the same initialisation along the same Brownian path, so the comparison is paired within a seed and the integrator is common to all of them. The discretisation was measured separately and is not what separates the arms: against an 800-step reference the strong error is 0.0190, 0.0087 and 0.0033 at 100, 200 and 400 steps, roughly halving per doubling, while the excess kurtosis of the generated sequences is stable at 2.00–2.05 across the whole ladder.

Table 4: Generation at $M = 2048$, sixteen seeds, every arm on the same Brownian path within a seed. The deficit is the paired difference in mean $\log p_{t_{\min}}$ per sequence against the arm driven by the exact score, in nats, with a seed bootstrap; the kurtosis column is |generated − reference|, the reference sample having excess kurtosis 1.936.

score driving the sampler	deficit vs exact (nats)		kurt. error	cov. error
exact (tree BP)	—		0.119	0.070
EM–BP ($C = 8$), fitted	+0.09	[+0.00, +0.19]	0.177	0.077
bidirectional GRU	+0.32	[+0.17, +0.46]	0.385	0.102
window head	+1.05	[+0.79, +1.31]	0.474	0.290
LMMSE (covariance twin)	+1.70	[+1.68, +1.71]	1.939	0.069
isotropic	+46.2		1.936	0.908

The ordering is the one the score errors predict, and the fitted kernel is nearest the exact score by a wide margin: its deficit is 0.09 nats against 0.32 for the best network and 1.70 for the covariance twin. The deficit is small but real — an earlier four-seed run suggested the fitted score matched the exact one, and sixteen seeds show it does not. The second-order score is the instructive failure. On the covariance error it is the best arm in the table, marginally ahead of the exact score, which is what Proposition 1 predicts. On the fourth cumulant it is the worst: it generates an essentially Gaussian chain with the right covariance. A generative model evaluated only on second-order distances would score it a success.

Where does the advantage stop? The structured estimator assumes a first-order Markov chain. Four Gaussian departures from that assumption were swept, each with a scalar strength ε and an exact score of its own, so every arm is scored against the truth of the perturbed model rather than against the chain.

Table 5: Misspecification at sixteen seeds: ratio of the window head's pooled error to EM–BP's, at zero strength and at the largest strength swept, $M = 2048$. Above 1 the misspecified chain estimator still wins. The break-even is reported as the bracket between the two grid points that straddle it, with the interpolated crossing in parentheses; we do not fit a crossing. The long-range row is the correction against the eight-seed draft.

mechanism	what it breaks	$\varepsilon = 0$	$\varepsilon_{\max}$	ratio	break-even
rank-one global latent	nothing structural	15.55	1.00	3.97	not reached
AR(2) second coefficient	the order	20.78	0.40	0.96	(0.3, 0.4), 0.38
long-range coupling	the precision band	14.11	0.40	1.00	not reached
site-varying ρ	homogeneity	14.22	0.14	0.98	(0.10, 0.14), 0.14

The pattern is interpretable, and it is not quite the one the eight-seed draft reported. A rank-one global latent is absorbed by the fitted correlation and never reverses the ordering,

because it does not break the conditional independence the estimator exploits; that stands. Two mechanisms that do break it — adding an order, and making the chain inhomogeneous — reverse it at a measurable strength, and the brackets are now tight.

Long-range coupling is the correction. At eight seeds its ratio fell below 1 at $\gamma = 0.4$ and was reported as a reversal near that point. At sixteen it does not: the paired log-ratio is $+0.004$ with interval $[-0.000, 0.009]$, a tie rather than a crossing, at both sample sizes. The eight-seed reading was inside its own noise. This also retires an inherited claim: the thesis reports the advantage reversing under long-range coupling near $\gamma \approx 0.1$, from a single draw per cell against an unscreened baseline. Against a screened baseline with sixteen seeds there is no reversal anywhere up to $\gamma = 0.4$.

What survives is the boundary the chapter lacked, stated more carefully: the advantage is not a general property of "having structure", it is specific to the structure being approximately right; breaking the order or the homogeneity costs it at a strength we can now name, and breaking the precision band or adding a global latent does not cost it within the range we swept. The two sample sizes also separate the two failure modes as predicted — at zero strength the EM–BP error halves from 0.019 to 0.010 between $M = 512$ and $M = 2048$, which is estimation; at the largest AR(2) strength it moves from 0.2135 to 0.2136, which is bias, and no amount of data will move it.

8 How far the results travel

Everything above is measured at one point of the model's parameter space: a Laplace innovation, $\rho = 0.85$, chain length 32. Four further runs ask whether that matters. Two of them strengthen the chapters; two of them require corrections, which are made here.

8.1 The capacity contrast holds in every regime

The capacity result is the load-bearing one, so it was re-run across five correlations $\rho \in \{0.3, 0.5, 0.7, 0.85, 0.95\}$ and three chain lengths $L \in \{16, 32, 64\}$, eight seeds each, everything else at the frozen protocol.

Table 6: Capacity across regimes: mean pooled relative score error of the Gaussian-innovation kernel ($C = 1$) and the mixture ($C = 8$), and their paired difference with a percentile bootstrap over seeds. Negative favours the mixture. Every interval excludes zero. $L = 64$ shown; the shorter chains agree.

ρ	t_*	$C = 1$	$C = 8$	paired difference	κ_4 of marginal
0.30	0.31	0.286	0.033	-0.253 $[-0.259, -0.246]$	2.50
0.50	0.55	0.266	0.035	-0.231 $[-0.239, -0.224]$	1.80
0.70	0.86	0.227	0.032	-0.195 $[-0.198, -0.192]$	1.03
0.85	1.22	0.167	0.028	-0.139 $[-0.142, -0.135]$	0.48
0.95	1.67	0.086	0.025	-0.061 $[-0.064, -0.058]$	0.15

Two things are worth having. First, the result is a property of the estimator and not of the corner: the mixture wins in all fifteen cells with the interval excluding zero, and the margin moves smoothly with ρ rather than appearing at one point. Second, and more useful to us, the sweep audits the apparatus at every cell. Proposition 1 says the $C = 1$ arm *must* coincide with the LMMSE of the true covariance; it does, to 0.002 in the median cell and 0.010 in the worst, and its fitted excess kurtosis is zero to three decimals everywhere. That is a check on the grid, the message passing, the EM and the scoring simultaneously, repeated at fifteen independent configurations, and it passes at all of them. The $C = 8$ arm recovers the innovation's true excess kurtosis of 3 at every correlation (2.85 to 3.40), with its spread across seeds widening from ± 0.33 at $\rho = 0.3$ to ± 1.38 at $\rho = 0.85$ — the shape gets harder to see, not differently

placed, which is the same information story from a third direction.

The margin shrinks monotonically with ρ, in the direction the theory gives: a strongly correlated chain has a nearly Gaussian marginal, its excess kurtosis falling as $3(1-\rho^2)/(1+\rho^2)$ (last column), so there is less fourth-order structure for capacity to buy. We report that as a direction and not as a rate. The pooled error runs over a schedule whose informative window itself moves with ρ, so its trend is not the exponent of the closure theorem, and fitting one to it would be over-reading.

8.2 The reference grid is not uniformly converged

Every number in this project is measured against tree-exact belief propagation on a grid of 401 points on $[-8, 8]$. That choice was validated once, at the frozen configuration; every other configuration we touch has inherited the validation without earning it. A sweep of five innovation families $\times$ four correlations $\times$ twelve levels $\times$ seven candidate grids against a (3201, 12) reference now measures it, and the result has a reassuring half and an uncomfortable half.

The reassuring half: *at the frozen configuration the operating grid is excellent.* The pooled drift of the posterior-mean field against the reference is 9.2×10^{-6}, the worst single level 3.7×10^{-5}, and 2.8×10^{-4} of the belief mass sits in the outermost 2% of the domain. Nothing in Sections 3–6 is in question.

The uncomfortable half: over the whole sweep, the worst-case drift is *flat in the number of grid points.* At half-width 8 it is 5.411×10^{-2} at 201 points and 5.425×10^{-2} at 1601 — refining by a factor of eight buys nothing — and it falls only when the domain widens, to 2.6×10^{-2} at (401, 10) and 3.0×10^{-3} at (801, 12). That is the signature of truncation, not of quadrature. It matters because "refinement" in this project, and in the thesis before it, has meant increasing the number of points, and that operation cannot fix this error.

The binding cell is Student-t_5 at $\rho = 0$: essentially all of its posterior mass falls outside $[-8, 8]$, and the drift is 5.4×10^{-2}. *That cell is reported*, in the phase appendix that orders the families by kurtosis at each correlation. Its entries are now flagged in the manuscript, marked in the figure, and excluded from the ordering claim; two recomputations at (801, 12) and (1601, 16) are queued to establish what they should be. No result in the chapters above depends on it — the chapters live at $\rho = 0.85$ with a Laplace innovation — but it is a reported number outside its validated envelope, and it should not have been.

8.3 The iteration budget, audited across every run

The capacity reversal was caused by a stopping rule. That made it worth asking, of every run we have, how often the validation-selected checkpoint is the *last* rung of the EM ladder — which is the signal that validation was still improving when the ladder ran out, and that a longer ladder might have selected differently.

The flagged claim. Section 4 states that the estimator attains the Cramér–Rao bound over the informative levels and leaves it past the crossover, at $t = 1.4$ and $t = 2$. The first half is safe: at every level up to $t = 1$, validation stops of its own accord well inside the ladder, so the budget is not binding and the efficiency result stands. The second half is not: at exactly the two levels where we report the estimator leaving the bound, it is also running out of iterations. The departure may be information or it may be budget, and this run cannot tell which. The manuscript now says so, and a control with the ladder extended sixfold is queued. Nothing else in the chapter depends on it.

The confounded run. The sample-complexity experiment was designed to escape the saturation problem by asking how much data buys back a fixed accuracy as the level rises — a

Table 7: Fraction of cells whose validation-selected checkpoint is the last rung of the 1200-iteration ladder. A high rate does not by itself invalidate a result; it means the result is a statement about a budget until something shows otherwise.

run	rate	reading
E02a (capacity)	up to 0.69 at $M = 2048$	*cleared*: the 400-iteration replication reproduces every overlapping cell to 0.001, so the estimates have plateaued well before the cap
E13 (regimes)	≤ 0.25; $C = 1$ never	no action
E08 (misspecification)	≈ 0.10	no action
E06 (learning)	0.12–0.38 for $t \leq 1$	*cleared* at the informative levels
	0.88 at $t = 1.4$, 0.75 at $t = 2$	**flagged**, see below
E12 (sample complexity)	1.00 at $t = 1$, $M \geq 4096$	**confounded**, see below

quantity that does not saturate. At $t = 0.6$ and $t = 0.8$ it worked. At $t = 1.0$ it did not: the error is flat and *non-monotone* in M, and the fitted kurtosis scatters between 0 and 3.4 across seeds with no concentration, because the EM is censored by the iteration cap in forty cells out of forty. At that level the experiment measured the optimiser's budget, not the sample size.

Two one-sided statements survive, and they are worth having because they come from opposite directions: the covariance coordinate's log-slope is 5.40 with interval [2.28, 7.72], which *excludes* the shape prediction of 8 and contains its own prediction of 4; and the shape coordinate fails to reach the accuracy threshold at $t = 1$ even at $M = 8192$, which forces its log-slope above 5.54 and *excludes* 4. The two coordinates are therefore separated, which is what the corollary asserts. The point estimates where a crossing exists sit on the predictions — 7.54 and 8.53 against 8, and 4.44 for the paired ratio against 4 — but their bootstrap intervals exclude nothing at all, and we do not offer them as evidence. A re-run with five levels, sixteen seeds and a ladder extended to 8000 is queued.

9 Status of the evidence, and what is still running

Refinement is complete and is as clean as this kind of check gets. Over the whole ladder — 401, 801 and 1601 grid points, half-widths 8 and 10 — every arm's error agrees to five decimal places and the reference field itself moves by at most 10^{-5}. No number in this report is an artefact of the quadrature at the frozen configuration. That sweep has since been extended to the other families and correlations, where the answer is less comfortable and is reported in Section 8.2.

Seven jobs are queued on the cluster against pre-registrations written before submission. Four are controls for the two problems above; one carries the quadrature envelope to the longer chains. The sixth and seventh are the one addition that is about strength rather than repair: **the information-matched comparison, re-run across three correlations and two chain lengths with the networks re-screened inside each regime**. That comparison is currently stated at a single point, and it is the claim a referee will press on hardest; the capacity result has now been shown to travel, and the comparison against networks has not.

Three infrastructure faults, for the record. Jobs were inheriting the login shell's 600-second CPU limit, which killed one run outright and left another holding an idle compute node for ten hours while the scheduler reported it running. Two drivers wrote their results only at the very end, so a timeout would have lost everything; both now write per cell and resume. And the site-varying misspecification mechanism is undefined at its largest registered strength. All three are fixed; the third is a registered amendment.

Table 8: Confirmatory runs. *Frozen* means every file in the output directory is pinned by SHA-256 in its registry entry, with the commit that produced it; a generator refuses to read an output that is not frozen or whose bytes have changed.

run	what it establishes	state
E01	order-by-order decay verified on the chain	frozen
E02a	capacity under equal stopping	frozen
E02b	capacity against innovation shape at fixed covariance	frozen
E03	six architectures, matched interface, length	frozen
E04	grid refinement on the fitted models	frozen
E05	generation under each score, paired noise	frozen
E06	learning through noise, against the information bound	frozen
E07	the second-order floor across correlation	frozen, one cell flagged
E08	misspecification and its break-even (16 seeds)	frozen
E09	infinite-data reference per capacity	frozen
E10	capacity in the thesis's own estimand	frozen
E13	the capacity contrast across ρ and L	frozen
E06b	the asymptotic exponents at $\rho = 0.5$	frozen, inconclusive
E11	the quadrature envelope across families and ρ	frozen; found the exclusion
E12	sample complexity by order	frozen, confounded
E06c	the same nine levels, ladder extended sixfold	queued
E07b, E07c	the family ladder at $(801, 12)$ and $(1601, 16)$	queued
E11b	the quadrature envelope at $L = 64$ and 128	queued
E12b	five levels, sixteen seeds, ladder to 8000	queued
E14	the matched comparison across ρ and L	queued

10 Decisions requested

1. **How should the capacity reversal be presented?** Either rewrite the capacity section of Chapter 10 around the new result and retain the old figure as a documented artefact, or keep both and make the diagnosis itself the finding. The second is more honest about how the error surfaced and costs roughly a page.

2. **Is the matched-interface comparison the headline, or a robustness check?** It is the stronger claim and it removes a scope condition the thesis carried. Promoting it changes the framing from "structure beats networks" to "structure beats networks at equal information, and the unmatched comparison understates it".

3. **Where does the efficiency result belong?** The estimator attaining the Cramér–Rao bound is what turns the two rates from an observation about EM into a statement about information. It reads naturally in Chapter 9 or in the theory chapter, and should be stated once with a reference from the other.

4. **How far should the Lean verification go?** Four algebraic constants are machine-checked. The remainder bounds are analysis and would be a substantial effort; the projection operator behind Proposition 1 is the one load-bearing step still in prose. Is the current scope the right stopping point?

5. **Do we pursue the asymptotic exponents, or claim the rates only where they are established?** Two attempts have now failed for two different reasons. The error-at-fixed-M route saturates: the window at $\rho = 0.5$ is three levels wide, too narrow to pin an exponent (Section 4). The sample-complexity route avoids saturation by construction but was confounded by the iteration budget at its highest level (Section 8.3); the re-run that removes the confound is queued, and if it lands we will have the exponents, and if

it does not we will have two one-sided statements that separate the coordinates without naming a rate. The fallback, which is what the write-up currently does, is to establish the rates exactly by computation and the estimator's efficiency by measurement, and to state the sample-complexity consequence only where the coordinate is identifiable. *Our recommendation: hold the claim at the fallback and let the re-run upgrade it if it can, rather than write the stronger version and retract it.*

6. **Does the misspecification boundary belong in Chapter 10 or in the conclusions?** It is the cleanest statement we have about when the method should *not* be used — breaking the order or the homogeneity costs the advantage at a strength we can name, while a global latent and a widened precision band do not cost it within the range swept. It reads as a limitation, but it is a measured one, which is unusual enough to be worth foregrounding.

7. **Do we wait for the regime sweep of the matched comparison before fixing the framing?** The headline factor of 7.3 is stated at one correlation and one chain length. The capacity result has now been shown to travel; this one has not been tested that way, and it is the claim a referee will press on. The run is queued and needs roughly two days of cluster time. *Our recommendation: yes* — the cost is small against the risk of building a framing on a single point, and the sweep can only strengthen or usefully qualify it.

8. **How should the quadrature exclusion be handled in the write-up?** One reported cell lies outside the validated envelope (Section 8.2). Three options: recompute and replace it silently once the queued runs land; recompute and report the movement as a numerical appendix; or keep the exclusion visible as a finding about the instrument, since "refining the grid does not fix a domain error" is a real and transferable observation about this class of method. *Our recommendation: the second* — honest, short, and it does not inflate a housekeeping fix into a contribution.

Every number above comes from a registered experiment whose output directory is pinned by SHA-256 in its registry entry, together with the commit that produced it and the seeds it used. The protocol shared by all arms is a single frozen object, so two arms differ only in the one factor each experiment varies. Runs were executed on the Bocconi HPC. Supporting material: the claim–evidence matrix, the negative-results record, the experiment registry and the reproducibility notes in the project repository.